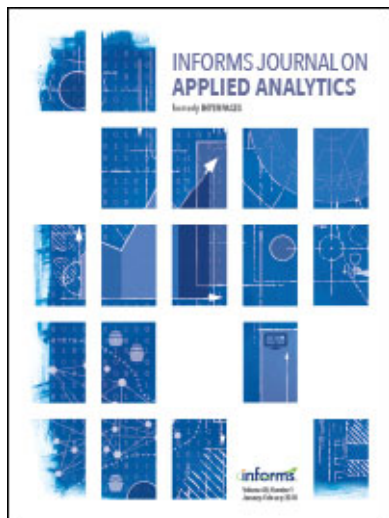




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JD.com Improves Delivery Networks by a Multiperiod Facility Location Model

Ningxuan Kang,^a Hao Shen,^{b,*} Ye Xu^c

^aJD.com, Beijing 100101, China; ^bSchool of Business, Renmin University of China, Beijing 100872, China; ^cBianlifeng, Beijing 100028, China

*Corresponding author

Contact: kangningxuan@jd.com,  <https://orcid.org/0000-0002-4637-1486> (NK); shenhao@rmbs.ruc.edu.cn,

 <https://orcid.org/0000-0002-1347-9394> (HS); ye.xu01@bianlifeng.com (YX)

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Abstract. Because delivery systems play an increasingly significant role in modern life, it is of strategic importance to design an efficient and adaptive delivery network. JD.com is the largest business-to-consumer e-commerce company by revenue in China and has built the largest amount of logistics infrastructure nationwide. In this study, we developed an optimization model to automatically make annual plans of the delivery stations that minimize operational costs. We built an optimization algorithm by adopting a multiperiod facility location model using mixed-integer linear programming (MILP) method and proposing applicable techniques to solve the model. The results demonstrate that our model generates improved plans compared with manually planning in terms of both operational cost and delivery distance. The tool described in this paper helps JD.com save on delivery costs and gain operational benefits in designing delivery networks.

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Keywords: dynamic facility location problem • delivery networks • mixed-integer linear programming

Introduction

Delivery systems are indubitably the key component of business logistics across industries, including e-commerce, mail and parcel services, and food delivery, to name but a few. With demand for delivery services now growing at a fast pace, it is of strategic importance to design efficient and adaptive delivery networks able to handle increasingly challenging situations.

JD.com is the largest business-to-customer e-commerce company by revenue in China. Compared with traditional retailing, delivery-to-door service is one of the greatest advantages of e-commerce practice, and JD.com has remained competitive among major e-commerce retailers for its leading-edge logistics system that enables high-quality delivery-to-door services. Specifically, as a self-operated online retailer, JD.com owns the largest amount of logistics infrastructure among all Chinese e-commerce companies. By the end of 2018, JD.com had 16 “Asia No. 1” logistics parks, 550 large warehouses, and 7,000 delivery stations (DSs), serving 99% of China’s population.

In recent years, JD.com has undertaken several market expansions and experienced a general increase in demand. As a result of this expansion, JD.com has realized that its existing delivery system will not be able to keep up with customer service targets in the future.

The mismatch between supply and demand is especially large during “6.18” and “11.11”—the two large sales promotions that JD.com runs annually on June 18th and November 11th. To guarantee service quality during these promotions, JD.com had to raise its shipping fees to cover transportation costs, a practice that has the potential to result in the loss of customers. Ultimately, JD.com has realized that a change in the structure of its delivery system is necessary to maintain customer service and lower operational costs.

JD.com’s distribution system consists of sortation centers (SCs), DSs, and connecting transportation. In changing its delivery systems, JD.com is embarking on adjusting citywide DS locations to improve the efficiency of its distribution system. This approach was chosen for two reasons. First, the locations of DSs, as the last layer in the system, have a pronounced impact on the efficiency of the entire system. Thus, an appropriate DS layout can reduce operational costs, improve delivery timeliness, and make logistics operations more convenient. Second, the DSs are leased on an annual basis; therefore, changes can be made when contracts expire.

Historically, JD.com’s DS location plans were made manually by staff in multiple departments from various regions of China. The planning procedure was time-consuming and susceptible to human error.

Therefore, JD.com wishes to find a more efficient way to make DS location plans that provide more convincing results.

The objective of the project was to develop a tool that is capable of automatically making annual plans for DS locations adjusted on a monthly basis that minimize total yearly operational costs. For practical use in JD.com, the tool required additional functions, including data preparation and results visualization. Nonetheless, this study mainly focused on the design of optimization models and solution methods.

We adopted a multiperiod facility location model to solve this problem. In doing so, we encountered several practical difficulties. First, JD.com's large size makes it impossible to rebuild its entire delivery network. Instead, the company wanted a gradual transition from its existing system to a more efficient one to guarantee its service quality. This led to diverse constraints that restricted how much we could change DS locations and coverages. Few existing multiperiod facility location models in the supply chain network design literature integrate all the practical restrictions faced by JD.com. This motivated us to propose a new model. Second, these restrictions, combined with the large size of JD.com's delivery networks, lead to a huge number of variables and constraints in the optimization model. We took Qingdao, a medium-sized city in China, as an example. There are 545 candidate DS locations and same-sized demand points in Qingdao. This constitution, together with JD.com's 12-period year, inspired an integer programming model with variables and constraints in the millions. For larger cities, such as Beijing and Shanghai, the size of the model becomes even larger and thus harder to solve. This challenge made it difficult to solve the model by directly using off-the-shelf solvers or existing methodologies in the literature. Therefore, we developed a specialized solution. Furthermore, the parameter estimation is a great challenge due to the complexity of the logistics system and the data issues encountered in practice.

The remainder of this paper is organized as follows. In the Literature Review section, we briefly review related literature on facility location models, applications, and existing solution methods, and we highlight our contributions. In the Model for Improving Delivery Networks section, we provide a detailed description of the problem settings and assumptions, as well as our modeling method. In the Solution Approach section, we propose several approaches for strengthening the model, and an efficient neighborhood-based heuristic for near-optimal solutions. In the Model Results section, we numerically test the efficiency and effectiveness of our solution approach, justify our model's assumptions, and demonstrate the benefits of our model over traditional manual plans. We then

briefly introduce the implementation of the DS location tool (in the Implementation section), the impacts of our project (in the Impact section), our pains and gains, and directions for future research (in the Lessons Learned, Challenges, and Future Plans section). Finally, we summarize this paper in the Summary section.

Literature Review

The facility location problem is an important category of operations research concerned with various models and solution techniques. We focused on the discrete optimization model, which basically consists of a collection of candidate facility locations and a set of demand points to be served. Decision makers need to determine where to locate the facilities and which demand points to serve for each facility to minimize network costs (which typically include transportation and fixed costs). We refer to Daskin (2013) for a detailed survey of the facility location models.

The model most closely related to our problem is the capacitated fixed-charge facility location model, in which the fixed cost of locating facilities as well as the capacity of facilities are taken into account. Applications include emergency facility locations (e.g., Duran et al. 2011, Aktaş et al. 2013, van den Berg et al. 2017), locomotive facility locations (e.g., Xie et al. 2016), and work center locations (e.g., Allen et al. 2017), to name but a few. A variety of exact algorithms and heuristics have been developed. Exact algorithms include branch-and-cut (Aardal 1998), Benders decomposition (Magnanti and Wong 1981), branch-and-price (Klose and Görtz 2007), and valid inequalities (Melkote and Daskin 2001). Heuristics and approximation algorithms include Lagrangian relaxation (Christofides and Beasley 1983), dual-based methods (Guignard and Spielberg 1979), tabu search (Ho 2015), and other heuristics (Jacobsen 1983).

An important extension of the capacitated facility location problem is the dynamic capacitated facility location problem (DCFLP). In a multiperiod setting, decision makers need to design a time-dependent plan to locate facilities with given capacities to meet demands in a given region. Arabani and Farahani (2012) thoroughly review the related models and solution methods. Meanwhile, Nickel and Saldanha da Gama (2015) focus on reviewing finite and discrete time horizon and discrete location space models, which are closely related to our work. Applications of this type include locations of new plants (Köksalan and Süral 1999) and clinical services across hospitals (Santibáñez et al. 2009). Associated solution methods involve dynamic programming (Wesolowsky and Truscott 1975), Lagrangian relaxation (Chardaire et al. 1996), branch-and-bound (Canel and Khumawala

1997), Benders decomposition (Castro et al. 2017), and heuristics (e.g., Dias et al. 2007, Jena et al. 2017, Tran et al. 2017).

This study departs from existing literature on DCFLP and its applications as follows. First, in addition to our focus on DS locations, we extend the literature by making the capacity of each DS a decision variable that is bounded from above by an exogenous limit. We also take into account the associated capacity-dependent fixed costs of new DSs. Second, our model has the capability to tackle diverse common practical restrictions on decision dynamics in the planning horizon. These restrictions include (i) upper-bounding the number of changes of both DS locations and block-DS assignments between two successive months, (ii) constraints associated with lease terms of existing and new DSs, (iii) limited-capacity expansion/reduction for each DS, and (iv) the monotonicity requirement of a monthly number of DSs across the year. Incorporating these practical considerations into the existing models proposed in the DCFLP literature will arguably destroy the structural properties on which the associated solutions build, thereby implying the necessity of developing a new model and a specialized solution. Third, for the convenience of practical operations, our model connects blocks served by the same DS. Modeling connectivity necessitates the introduction of a considerable amount of decision variables and constraints such that the model becomes intractable even for small instances. To the best of our knowledge, there is no DCFLP model that systematically integrates all these considerations, which substantially complicate the model and demand a specialized methodology.

Ultimately, we present a customized reformulation strategy together with valid inequalities to effectively reduce the problem scale and strengthen the model. Also, by exploiting the properties of our real-world problem, we offer a novel and efficient neighborhood-based heuristic, leveraging a nested warm-start algorithm for near-optimal solutions.

Model for Improving Delivery Networks

Basic Settings

A citywide distribution system involves two types of logistics infrastructure: an SC and a DS. An SC receives packages from warehouses or other SCs, sorts them by their destinations, and sends them to corresponding DSs. A DS receives packages from SCs and eventually delivers them to end customers. DS staff consist of a station manager and several delivery men. JD.com geographically partitions a city into blocks of particular shapes, sizes, and boundaries, and one delivery man typically delivers packages to a single block. Each DS delivers

packages to its surrounding blocks, and each block is assigned to exactly one DS for operational simplicity.

We modeled the citywide delivery network as a directed tripartite graph. Figure 1 illustrates the structure of this graph. There are three layers of vertices in the graph: SCs, DSs, and blocks (demand points). The arcs from SCs to DSs, and from DSs to blocks, represent the corresponding transportation. We assumed that DSs can only be located in blocks, and that a block can accommodate one DS at most. We adjusted DS locations and the assignments of blocks to DSs in our model to produce new plans. It is assumed that the locations of SCs in the city are given and remain fixed, as relocating SCs is costly, and candidate locations for SCs are usually limited. Furthermore, we omitted the transportation cost between SCs and DSs, as it is negligible compared with the transportation cost between DSs and blocks, and changes very little under different DS locations (we provide numerical justification for this assumption in the Model Results section).

As we noted, all of the DS sites are leased and renewed on an annual basis. Once a new DS is set up, it cannot be removed until the end of the lease. An existing DS keeps its size if renewed, whereas a new DS chooses its size from a reasonable range (estimated from practical experience) and stays constant once decided. The capacity of a DS refers to the maximum number of packages it can deliver in a month and is usually related to its size. In addition, the fixed cost associated with setting up a new DS also depends on its size. We assume that the capacity and associated fixed cost of a new DS are both linear functions of its size.

Practical Restrictions

We introduced several practical restrictions based on the operational practice of JD.com. Some of these restrictions may be optional for certain cities, but we integrated all practical restrictions into our model to increase the acceptance of the DS location plan and lower the risk in its launch. We organized these restrictions into three groups.

The first group of restrictions was regarding DS location changes. Changing the location of a DS refers to either opening a new station or moving an existing station to a different location, both of which require efforts to find new sites. There are two types of restrictions: (1) the upper bound on the number of DS location changes every month and (2) the monotonicity requirement of the total DS number (as a result of DS location changes) over months. We applied the upper-bound restriction for three reasons. First, during peak months, such as June and November (when the “6.18” and “11.11” sales promotions take place), no changes can be made to the delivery network, because they may impair service quality. Second, during

the holiday season, such as January and February (Chinese New Year), staff usually take vacations, making the operation team shorthanded and rendering any changes to delivery networks difficult. Hence, we avoided any changes during these two months as well. Third, even during a regular month, it is still challenging for the operation team to launch a plan if too many location changes are recommended in a given month. Therefore, we set an upper bound on the number of location changes that are permitted in each regular month. The monotonicity restriction is to avoid fluctuations in the number of DSs due to seasonal shifts in market demand, which are undesirable for human resource management. In cities with surplus DSs and low DS use rates, we let the number of DSs decrease over months to remove excess operational costs; meanwhile, in cities with insufficient DSs, we let the number of DSs ramp up over months to increase delivery capacity.

The second group of restrictions was on block assignment changes. A “block assignment change” signifies the reassignment of a block to a different DS. If a block previously assigned to DS 1 is reassigned to DS 2, then the delivery man responsible for this block has to change his personnel organization to DS 2, or a delivery man in DS 2 has to take over the delivery job of this block. To avoid the frequent reallocation of human resources, we put an upper bound on the number of assignment changes per block during the planning horizon.

The third group of restrictions was related to the connectivity requirement. First, it is important to define “block adjacency” and “connectivity.” We call two blocks “adjacent” if they share at least one side or one vertex. As shown in Figure 2, block A is adjacent to block B and block C, whereas block B and block C are not adjacent to each other. We define a “path” of blocks as a sequence of blocks in which successive pairs of blocks are adjacent. We call a set of blocks “connected” if, for any two blocks in the set, there exists a path between them. We required all blocks assigned to the same DS to be connected in order to benefit from operational flexibility. In Figure 3, we provide an example to illustrate connectivity and its benefit. There are two DSs in this area, DS 1 and DS 2, and each covers several blocks. In Figure 3(a), the blocks covered by DS 1 are not connected, since block B is not adjacent to any other block covered by DS 1. It is difficult for block B to receive assistance from DS 1 if its delivery man takes sick leave or if there is a surge in the number of packages, because the other delivery men from DS 1 have to go across some blocks covered by DS 2 in order to lend a hand. On the contrary, in Figure 3(b), by reassigning block A from DS 2 to DS 1, all blocks covered by DS 1 are connected, so that

delivery men in any other block of DS 1 can easily offer assistance to block B.

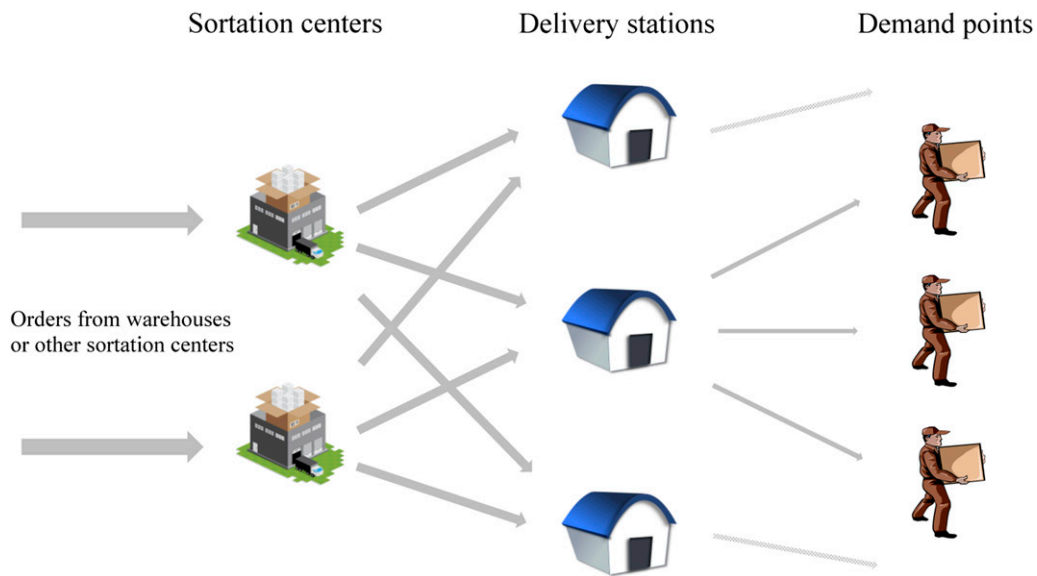
Parameter Preparations

Several types of parameters were prepared for the model. The first type of parameter was imported by the operation team based on practical circumstances, including lease durations of DSs, the maximum number of location and assignment changes they are willing to make, and the monthly number of packages deliverable to each block in the planning year. The monthly demand of each block in the planning year was deterministic in our model, which is consistent with the previous manual planning practices of JD.com. The number of packages in each future month is a preset objective for the operations department of JD.com to achieve in the forthcoming year. If in the future the realized demand in a certain month has not yet reached the preset objective, then the operations department must adjust its operational strategy by using promotions to stimulate sales and reach the objective. Thus, we can use these preset monthly numbers of packages in each block as deterministic demands in the model without deviating too much from realized demands.

The second type of parameter was estimated based on existing JD.com data. We estimated the capacity of each DS as a linear function of its size; meanwhile, we estimated its fixed cost (including rent, operational cost, wages, etc.) based on its location and size. We fit the delivery cost of a package with a linear function of the delivery distance. This way of calculating delivery cost is based on the operational practices of JD.com: delivery staff are paid for every package they deliver successfully; this payment represents the delivery cost incurred to the company per package—the routing costs are mostly human costs. More specifically, JD.com’s payment rate per package depends on the distance between the DS and the delivery address of a package. Thus, for each city, we collected payment rate (unit delivery cost) data as well as associated distances; we found that a linear function of distance (estimated by linear regression) was sufficiently capable of characterizing the unit delivery cost.

The third type of parameter was obtained from external sources. For example, with the help of map service providers, we obtained the delivery distances between DSs and blocks (approximated by the driving distances between DSs and the geometric centers of blocks). By geometrically analyzing block boundaries, we identified adjacent blocks and checked whether any blocks were isolated by rivers or freeways. When the latter occurred, we fixed the data issue by setting the isolated block as adjacent to its nearest neighbor.

Figure 1. (Color online) A Tripartite Graph Configuration of the Citywide Distribution Network of JD.com



Note. The distribution network of JD.com consists of three layers of vertices, namely, sortation centers (SCs), delivery stations (DSs), and blocks (represented by demand points), as well as transportation arcs between SCs and DSs, and between DSs and blocks.

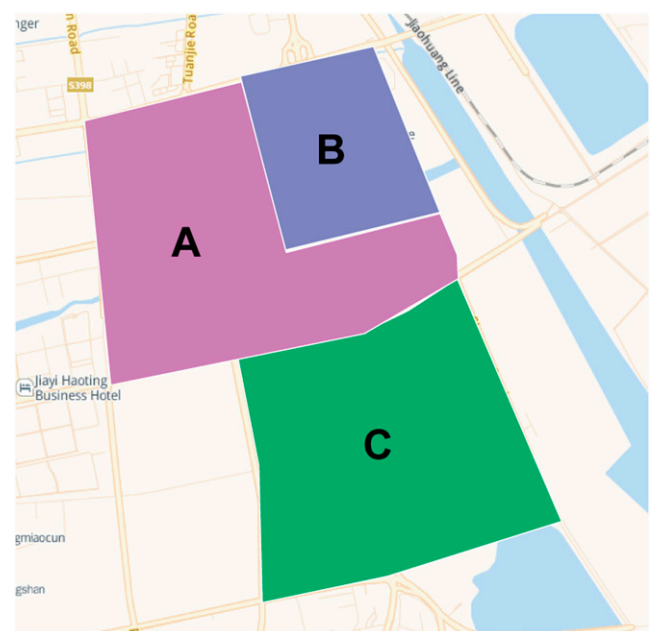
Formulation

Our goal is to generate an annual plan for every city to adjust delivery networks on a monthly basis. We formulated this problem as a multiperiod facility location model. The objective of the model was to minimize the total cost of the delivery network, which consists of the fixed costs of DSs and the delivery costs from DSs to blocks. The decision variables of the model included the location and the size of each DS, and the monthly assignment of each block to a DS. In addition, we incorporated some intermediate variables to represent fixed costs, assignment changes, and connectivity. Thus, the decision variables were binary or continuous, and the objective function was formulated as a linear function of the decision variables (refer to Appendix A for mathematical details).

In sum, we considered four groups of constraints: (i) initial conditions, (ii) facility location-allocation constraints, (iii) practical restrictions, and (iv) connectivity requirements. Group (i) initializes the locations of existing DSs and the assignments of blocks at the beginning of the planning horizon (see Constraints (A.2)–(A.5) and (A.17) in Appendix A). Group (ii) guarantees that each block is assigned to one DS and that the package quantity assigned to each DS does not exceed its capacity (see Constraints (A.18)–(A.20) in Appendix A). Group (iii) describes the lease restrictions, and the limits on the number of location and assignment changes (see Constraints (A.6)–(A.16) and (A.21)–(A.23) in Appendix A). Group (iv) ensures that blocks assigned to the same DS are connected (see

Constraints (A.24)–(A.29) in Appendix A). In addition, Constraints (A.30) listed in Appendix A reveal the calculations for the fixed costs of DSs used in the objective function.

Figure 2. (Color online) Example of Block Adjacency



Notes. This figure illustrates three blocks as an example to explain the concept of block adjacency. In the figure, block A is adjacent to block B and block C, whereas block B and block C are not adjacent.

Solution Approach

Practical restrictions, including connectivity constraints, made the model difficult to solve directly. To address this problem, we first employed an alternative formulation of the connectivity constraints to reduce the number of both decision variables and constraints. Next, we strengthened the model by using several valid inequalities and tighter bounds on decision variables. The related mathematical details are provided in Appendix B. In addition, we also used a neighborhood-based heuristic, leveraging a nested warm-start algorithm to enhance the solution efficiency for full-scale delivery networks. We briefly introduce the details of the heuristic method.

As practical experience indicates, blocks are more likely to be designated to a nearby DS rather than a remote one. Hence, we sought near-optimal solutions by defining the $\tau\%$ -neighborhood-restricted problem, which limited the assignment of each block to DSs within its top $\tau\%$ nearest blocks. We also used the parameter τ as the relative size of the restricted DS neighborhood. Helpful to recall here is that the original (unrestricted) problem was a special case of the neighborhood-restricted problem, that is, the 100%-neighborhood-restricted problem, where the neighborhood of each DS is the set of all blocks. We can tune the size of each DS neighborhood to balance optimality and computational efficiency. In the Model Results section, we show that fine-tuning the size of a DS neighborhood (usually its top 10% nearest blocks) reduces the complexity of the problem, thereby tremendously boosting computational efficiency without losing significant optimality.

We also designed a nested warm-start mechanism to accelerate the computation process. By analyzing

the solution procedure, we found that the optimizer meaningfully explored an initial feasible solution. To tackle this, we first defined a sequence of “nested” neighborhood-restricted problems: for every two successive problems in this sequence, the neighborhood of each DS in the former problem was limited to a subset of the neighborhood of the same DS in the latter problem. For instance, three problems, namely, the 5%-, 6%-, and 7%-neighborhood-restricted problems, constitute a sequence of nested neighborhood-restricted problems. We then solved the first problem in the sequence for an optimal solution (which is feasible for the second problem) and used this solution as a warm-start (or initial feasible) solution to solve the second problem. We repeated this procedure until the objective neighborhood-restricted problem was solved. We summarize this algorithm in Algorithm B.1 in Appendix B.

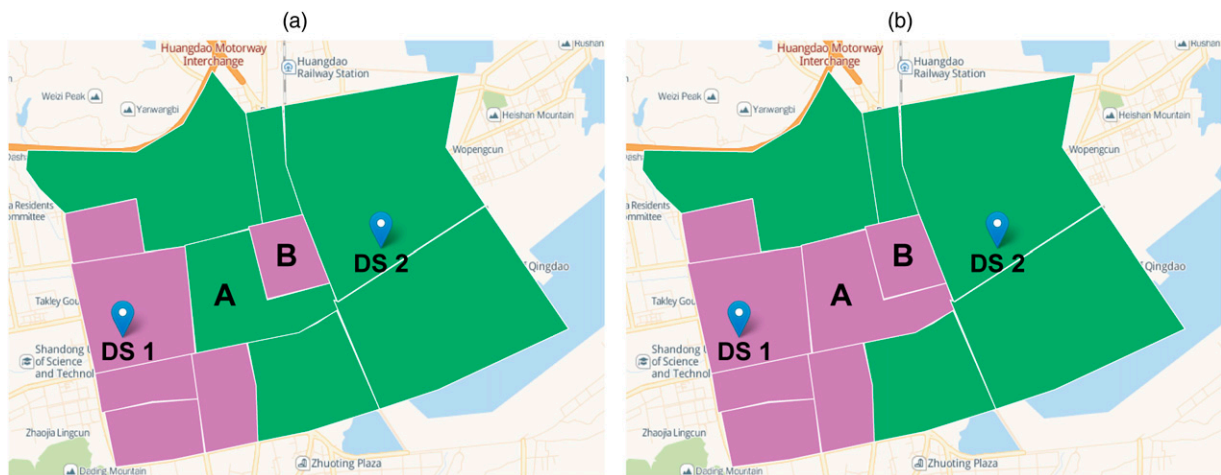
Model Results

We take the annual planning in Qingdao as an example to show the effectiveness of our model. We also provide a supplementary case study on the delivery network in Dalian with similar results in Appendix C.

Computation Efficiency

Qingdao is a medium-sized city in the east of Shandong Province on the coast of China’s Yellow Sea. It contained 51 DSs and 545 blocks at the beginning of 2018. The R^2 of the linear model between the delivery cost per package and the delivery distance for Qingdao in 2018 was 0.909, which suggests that our linear regression model fits the data well. We solved our model to generate an annual plan for Qingdao using Gurobi 8.1.0 on a CentOS Linux 7.4.1708 operating

Figure 3. (Color online) Two Different Plans of Block Assignments for Two Delivery Stations



Notes. The figures illustrate two different plans of block assignments for two delivery stations (DS): DS1 and DS2. In the left figure, the blocks assigned to DS1 are not connected, because block B is not adjacent to any other block covered by DS1. In the right figure, the blocks assigned to each of DS1 and DS2 are connected, which satisfies JD.com’s connectivity requirements.

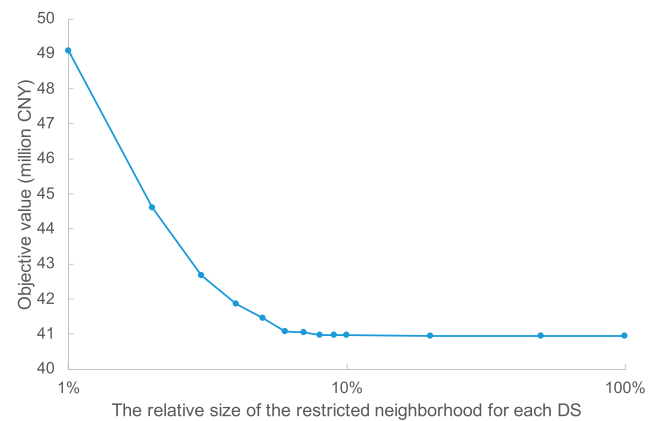
system equipped with two Intel Xeon E5-2698 v3 processors and 256 GB RAM. We selected τ from the set $\{1, 2, \dots, 10, 20, 50, 100\}$ and sequentially solved the associated nested τ %-neighborhood-restricted problems. We terminated the solution procedures with a 0.30% optimality gap for each neighborhood-restricted problem. We summarize the computation performance in Table 1 and the objective value in million Chinese yuan (CNY) in Figure 4 for each neighborhood-restricted problem. The heuristic gap for a τ %-neighborhood-restricted problem in Table 1 is defined as the ratio of the difference between the optimal values of this problem and the unrestricted problem ($\tau = 100$) to the optimal value of the unrestricted problem.

The results in Table 1 and Figure 4 show that as the relative size of a restricted DS neighborhood grows up to 10%, the objective value (or the total cost) reduces significantly within a reasonable computation time frame. However, when the relative size is beyond 10%, a considerable amount of computation effort still has yet to be paid, and little cost improvement is gained. Therefore, solving the 10%-neighborhood-restricted problem by using our proposed nested warm-start solution method can substantially save computation time without losing significant optimality. In the Dalian case in Appendix C, 10% remains a good relative threshold size, effectively balancing optimality and computational efficiency (see Table C.1 and Figure C.1 for detailed results).

Accuracy of Omitting Transportation Cost Between SCs and DSs

For the Qingdao case, we numerically justified the assumption that transportation cost between SCs and DSs can be omitted in the Model for Improving Delivery Networks section. We compared our models with

Figure 4. (Color online) Changes to the Objective Value of the Neighborhood-Restricted Problem



Note. This figure illustrates how the objective value (or the total cost) of the neighborhood-restricted problem changes as the relative size of the restricted neighborhood varies in the solution procedure for the Qingdao instance.

a model that incorporates the transportation cost between SCs and DSs, and that was therefore a trivial extension of our model. Table 2 shows the monthly results of the transportation cost (TrC) between SCs and DSs, and the total cost (ToC) of the annual plans generated by the two models, namely, the model considering TrC and the one omitting TrC. The results demonstrate that TrC only accounts for a small proportion (up to 7.1%) of the total cost and that the relative difference of TrC incurred by the two plans is also low (under 4%). These findings led to a tiny relative change (0.1%) in the annual total cost between the two models. Therefore, omitting the transportation cost between SCs and DSs does not materially impact total cost.

Table 1. Computation Performance Results for the Neighborhood-Based Heuristic with the Nested Warm-Start Algorithm in the JD.com Network in Qingdao

Relative size, τ^a	Marginal time (seconds) ^b	Accumulative time (seconds) ^c	Heuristic gap (%)
1	34.61	34.61	19.911
2	46.93	81.54	8.985
3	131.25	212.79	4.210
4	8,591.14	8,803.93	2.250
5	28,803.17	37,607.10	1.251
6	1,396.80	39,003.90	0.338
7	1,856.73	40,860.63	0.259
8	6,068.74	46,929.37	0.067
9	1,087.30	48,016.67	0.063
10	1,859.24	49,875.91	0.062
20	10,437.88	60,313.79	0.014
50	9,792.06	70,105.85	0.002
100	18,335.57	88,441.42	0.000

^aRelative size of restricted DS neighborhood.

^bThe computation time of solving the current problem by using the optimal solution of the former problem as the warm-start solution.

^cThe total computation time of solving the current problem.

Table 2. Cost Comparison Between Models Considering and Omitting the Transportation Cost Between SCs and DSs in Qingdao in 2018

Month	Considering TrC			Omitting TrC			Relative change (%)	
	TrC (CNY)	ToC (CNY)	$\frac{\text{TrC}}{\text{ToC}}$ (%)	TrC (CNY)	ToC (CNY)	$\frac{\text{TrC}}{\text{ToC}}$ (%)	TrC	ToC
January	242,676	3,866,730	6.276	242,676	3,866,730	6.276	0.000	0.000
February	242,676	3,424,533	7.086	242,676	3,424,533	7.086	0.000	0.000
March	199,884	3,623,423	5.516	199,884	3,622,991	5.517	0.000	−0.012
April	190,380	3,431,312	5.548	190,380	3,430,881	5.549	0.000	−0.013
May	171,310	3,457,209	4.955	171,310	3,458,018	4.954	0.000	0.023
June	171,310	3,568,631	4.800	171,310	3,569,498	4.799	0.000	0.024
July	166,576	3,531,540	4.717	166,576	3,532,407	4.716	0.000	0.025
August	157,037	3,492,075	4.497	161,781	3,503,088	4.618	3.021	0.315
September	133,297	3,428,861	3.887	138,056	3,438,808	4.015	3.570	0.290
October	128,469	3,439,456	3.735	133,229	3,448,329	3.864	3.705	0.258
November	128,469	3,871,748	3.318	133,229	3,877,987	3.436	3.705	0.161
December	133,239	3,876,106	3.437	137,999	3,883,784	3.553	3.573	0.198
Total	2,065,323	43,011,625	4.802	2,089,106	43,057,055	4.852	1.152	0.106

Note. CNY, Chinese yuan; ToC, total cost; TrC, transportation cost.

Benefits

We compared the plan generated by our model with the manual plan made by the business team. Table 3 summarizes the costs, DS capacity use rate, and delivery distance of both plans. Our model saved 8.08% of the total cost compared with the manual plan. Not surprisingly, most of these cost-savings were due to the reduction in DS fixed costs, as the number of DSs was almost cut into half by our plan. As a result, the average DS use rate had an 11.56 percentage point (pp) increase. This implies that the initial delivery network had surplus DSs. Although our model reduced the number of DSs, it did not increase the delivery distance; on the contrary, the average delivery distance was shortened from 5.53 to 5.33 kilometers (km) (−3.61%), which significantly improved customer experience.

To better benchmark our model results, we also ran a model without initial conditions, namely, the initial DS locations and the initial block-DS assignments, for Qingdao in 2018. That being said, the benchmark model selected the best DS locations and block assignments from the first month without considering existing DSs and thus provided a theoretical lower bound on the total cost. This bound, although it cannot be reached in practice, indicates the maximum potential for cost savings. We compared our model plan with the benchmark plan in the last two columns of Table 3. Our model resulted in improvements equal to nearly

50% of its potential (8.08%, with a possible 16.45% reduction in total cost) and recommended a DS number very close to that in the benchmark plan (29 vs. 28).

We plot the DS locations in Qingdao at the end of the planning horizon (December 2018) recommended by the benchmark plan, our model plan, and the original manual plan in Figure 5. The blue pins denote DS locations, whereas the white, gray, and black blocks in Figure 5, (b) and (c), can be interpreted as follows: Compared with the benchmark plan in Figure 5(a), white blocks are covered by the same DS without location change, gray blocks are covered by the same DS that is moved to another block nearby, and black blocks are covered by a new DS. The maps illustrate that our model plan is more similar to the benchmark plan than it is to the manual plan, with regard to the network configuration. Nonetheless, the benchmark plan suggests 26 DS changes at once in January, whereas our model recommends an easier-to-implement plan by distributing 17 DS changes to seven months, each of which is assigned up to seven DS changes.

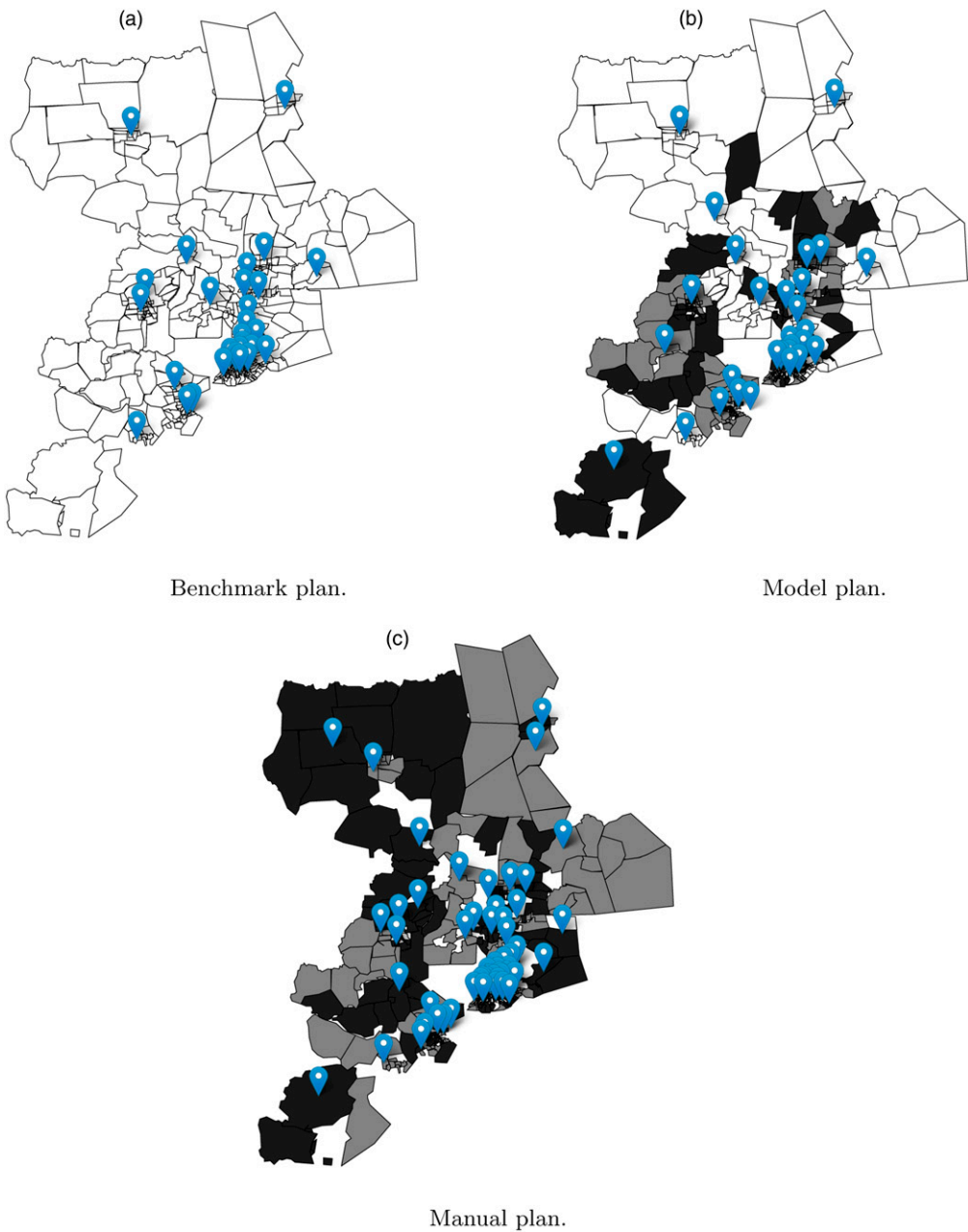
Implementation

Our model was embedded in a new DS location planning tool designed for the annual planning of JD.com's delivery system. We contributed to the tool by developing its core optimization program, including the DS location model and the solution algorithm, and by

Table 3. Plan Comparison for Qingdao in 2018

	Manual plan	Model plan (% change)	Benchmark plan (% change)
Total cost (CNY)	44,567,733	40,967,948 (−8.08%)	37,235,276 (−16.45%)
Fixed cost (CNY)	14,270,108	10,927,427 (−23.42%)	7,333,800 (−48.61%)
Delivery cost (CNY)	30,297,625	30,040,521 (−0.85%)	29,901,476 (−1.31%)
DS number in the last month	54	29 (−46.30%)	28 (−48.15%)
DS capacity use rate	37.14%	48.70% (11.56 pp)	74.18% (37.04 pp)
Average delivery distance (km)	5.53	5.33 (−3.61%)	5.22 (−5.58%)

Figure 5. (Color online) Comparison of the DS Locations and Block Assignments



Notes. The figures show a comparison of the DS locations and the block assignments of the benchmark plan (a), our model plan (b), and JD.com’s previous manual plan (c) for Qingdao in 2018. The blue pins denote DS locations; the white, gray, and black blocks in the middle and right figures can be interpreted as follows: compared with the left figure, white blocks are covered by the same DS without location change, gray blocks are covered by the same DS, which is moved to another block nearby, and black blocks are covered by a new DS.

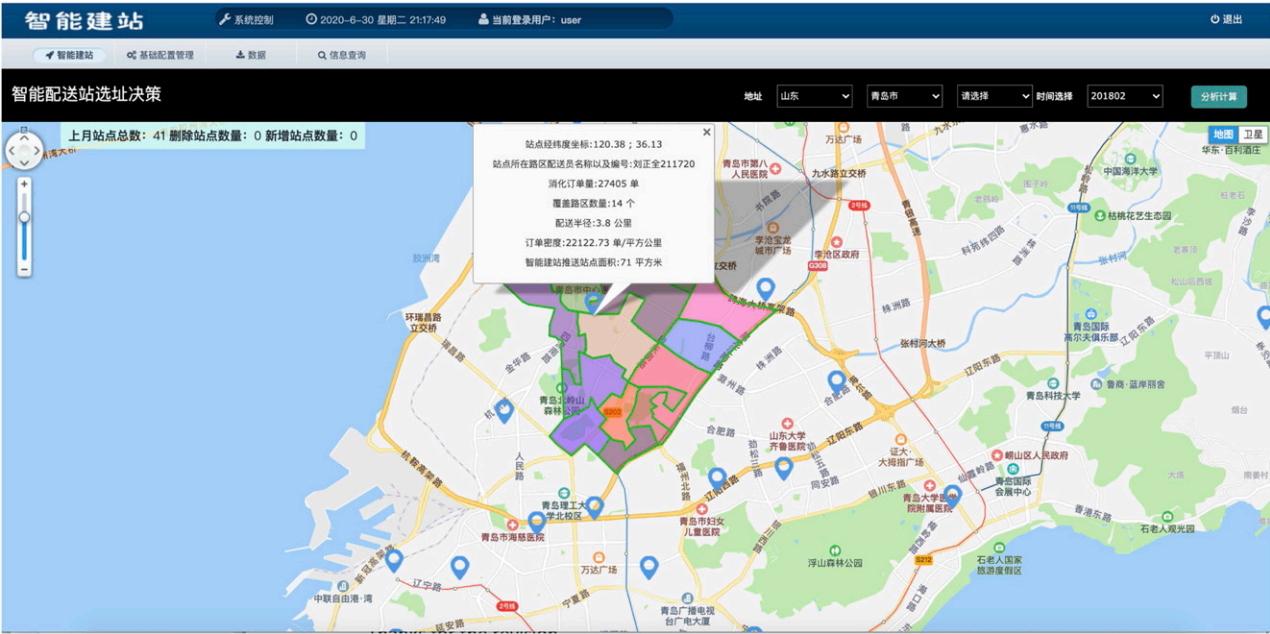
designing both the architecture and the visual interface of the tool. In addition, we also helped the engineering team with the software coding and testing processes. We briefly introduce the main functions of the tool.

To generate a plan for a city, the DS location planning tool first extracts historical operational and geographical data from JD.com’s Big Data Platform and generates input data for the model. The optimization program then builds the optimization model with

input data and solves for the optimal solution for each city. After the solution procedure is complete, the tool converts the solution into a plan and displays it on a visual interface, as shown in Figure 6.

In the visual interface, all of the DS locations and blocks are shown on a map, and blocks covered by the same DS have the same color. Once the user clicks on a DS mark, its basic information pops up and the blocks covered by that DS are highlighted on the map

Figure 6. (Color online) Visual Interface of JD.com’s DS Location Planning Tool



Notes. The screenshot shows the visual interface of JD.com’s DS Location Planning Tool. The screenshot displays the (colored) blocks assigned to a delivery station in our model plan for Qingdao in February 2018.

(Figure 6). Users can conveniently query DS locations and block assignments through this visual interface. The map view also enables a comparison of the model plan and the benchmark plan (discussed in the Benefits section). Furthermore, we designed a module to export the plan into spreadsheets, making it easier for the operation team to share, analyze, and modify the results.

JD.com selected 16 major cities, covering more than 650 million packages in China, as pilot cities for our project. The operation team in charge of these pilot cities can download and use the recommended plan to guide the practical design of their delivery networks, including opening, closing, and relocating DSs. The prototype of the DS location planning tool has been endorsed by the operation team of JD Logistics.

Impact

We generated 2018 annual delivery network plans for the 16 pilot cities with the help of the model described in this paper. The results show that the plans generated by our model have great economic impacts. Compared with manual plans, it was estimated that our plans could save over 82 million yuan in DS deployment and delivery services, 4.2% of the total cost of DS networks in 2018.

Moreover, our project also demonstrated noneconomic impacts. First, our plans outperformed the manual plans in terms of customer experience. Our models showed that the average delivery distance per

package from DSs to customer could be reduced by 20.9%, from 3.83 to 3.03 km. This reduction would then significantly improve the timeliness of delivery and thus customer experience as well as network efficiency. Second, the DS Location Planning Tool increased the employee efficiency of JD.com. Traditionally, plans were made manually by staff from many departments, and it usually took at least a month to manually make an annual plan for a medium-sized city; for larger cities, the planning process was even more time-consuming. By contrast, our model, embedded in the DS location planning tool, provides an automatic mechanism for the company to make DS location plans only within hours. Last but not least, our project, which successfully applied operations research techniques, has been recognized throughout the company and has made other departments interested in the critical value of automated and optimized decision support.

Lessons Learned, Challenges, and Future Plans

We devoted great efforts to persuading the operation team at JD.com to accept and use the DS location planning tool for their annual plans of delivery networks. At the beginning of the project, we spent huge amounts of time communicating with potential users of the tool from diverse departments in the corporation and understanding their needs, requirements, and expectations. Throughout this project, we constantly

received feedback from the operation team and accordingly modified our model. Several times, constraints were added in response to the new practical restrictions, and these restraints fundamentally complicated the model, disabled the original solution methods, and motivated us to develop different solutions. After finishing the prototype of the DS location planning tool, we collected numerous manual plans of different cities, as well as associated data, and conducted extensive numerical experiments to demonstrate the effectiveness and the efficiency of our model. We also incorporated several extended and optional functions in the tool to tackle different practical restrictions that may arise in the future.

Nonetheless, several practical challenges complicate the implementation of our recommended plan. One principal challenge is that appropriate housing resources are not always available in designated blocks. An appropriate location for a DS should be (i) ground level with an independent entrance and a sufficiently large parking space and (ii) in a neighborhood without heavy traffic. In addition, the rental price in certain blocks can be much higher than expected. Recognizing this challenge, our team plans to continue research to optimize implementation. First, we plan to add access to detailed and real-time information of housing resources, including prices, locations, sizes, and so on, from third-party platforms to the DS location planning tool. Meanwhile, we will incorporate this information into our decision model and enable the tool to recommend not only good blocks for particular DSs but also the appropriate housing resources within each block. Second, we plan to collect historical traffic data and extend the tool by providing traffic heat maps to help the operation team efficiently select suitable neighborhoods as DS locations.

Another challenge is the difficulty in updating our recommended plan frequently and responsively as a demand surge or inconsistency between the implementation and our recommendation is observed or expected. Originally, to cope with demand or implementation uncertainties, the DS planning tool was designed for use at any time point for any planning horizon, as long as data were available. However, frequently and responsively updating the recommended plan is difficult in practice for two reasons. First, frequent plan changes can impair service quality and are difficult to implement, because of, for example, the housing resource issue. Second, in the practice of JD.com, it takes over a month to enter a leasing/renewal decision into the system, because it is a manual process done on an ad hoc basis. Given the long feedback loop of offline operations, it is almost impossible to run the planning tool with an accurate new initial condition and update the plan responsively. These issues, enclosed in the feedback that we received from

the operation team in the first quarter of 2018, led to a partial implementation of our recommended annual plans in 2018. For example, the realized annual demand of Qingdao in 2018 was observed to be 40% higher than expected due to a significant increase in third-party waybills and an aggressive expansion of the fresh food market. As a consequence, the actual network at the end of 2018 deviated from the model plan; some DSs that the model plan recommended to be closed were kept open, and a few new DSs were set up at unplanned areas, resulting in higher operational costs.

To overcome this challenge and to shorten the offline feedback loop, we have been working with JD.com's business and engineering teams to build a standard data collection procedure and an easier-to-use interface for the data collection system. In addition, to strengthen the adaptability of the DS Location Planning Tool, we plan to extend it to allow for manual changes and provide what-if analyses for different situations.

Despite the challenges, the results generated by our model *instructed* the operation team to design new yearly plans. For example, compared with the original manual plan, the model results advised fewer stations in remote regions, so the operation team then removed redundant stations in these areas. Another optimization step is the timing of opening new DSs. The manual plans tended to open new DSs at the beginning of a year. Our model results recommended that new DSs were opened according to package quantity of each month. These suggestions were all adopted by the operation team in the delivery system design for pilot cities in 2018.

Summary

After experiencing several market expansions, JD.com now faces logistical challenges; its system must be revised. More specifically, today, quick-shifting demand, redundant infrastructure, and market competition heavily burden its logistics system. The manual adjustments of the delivery network JD.com has traditionally used are now inefficient and costly. Accordingly, we leveraged our expertise in operations research to formulate the planning problem as a multiperiod facility location model and developed an automatic location planning tool to improve JD.com's delivery networks. Our model minimizes the total cost of the network and accounts for a variety of practical constraints. More specifically, we developed an efficient neighborhood-based approximation model and incorporated a warm-start mechanism to accelerate the solution. Numerical tests verified our model's effectiveness. Compared with JD.com's manual plans, our model generates more efficient networks with shorter delivery distances, higher DS use rates, and lower operational costs. We embedded the model into an automatic planning tool that enables JD.com to efficiently generate an optimal plan for every city in China.

Our work follows the “customer first” principle, one of the core values of JD.com. An efficient delivery network makes it possible for customers to pay less and enjoy better services at the same time. On one hand, shorter delivery distances enable the company to provide faster and more reliable delivery. On the other hand, lower operational costs enable the company to offer lower prices to customers in the long term. In the future, JD.com will extend its implementation of this tool to all cities to achieve greater savings and bring even more value to customers.

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Appendix A. Mathematical Model

In this section, we present the notation of the model, as well as its mathematical details. We applied a mixed linear integer programming (MILP) method to generate the yearly plan of delivery station locations. Following is a summary of the notation.

Indices

- $j \in J$: index of either a block or a delivery station located at this block; J is the block set
- $j \in \hat{J}$: index of a block without a delivery station, $\hat{J} \subseteq J$
- $j \in \bar{J}$: index of a block that cannot be selected for DS, $\bar{J} \subseteq J$
- $t \in T$: index of a period; $T = \{1, 2, \dots, 12\}$ for a yearly plan

Parameters

- d_{jk} : distance in kilometers between block j and block k , $j, k \in J$
- α_d : constant term in unit delivery cost function of packages (in Chinese renminbi [RMB])
- β_d : coefficient of delivery distance in unit delivery cost function of packages (in CNY/km)
- n : limits on the number of assignment changes for each block over a year
- n_t : limits on the number of location changes of all delivery stations during month t ; a location change includes adding a new station or relocating an old station, $t \in T$
- n_t^c : limits on the number of assignment changes for all blocks during month t , $t \in T$
- t^{start} : first month when adjustments are allowed, $t^{start} = \min\{t \in T | n_t > 1\}$
- m_t : the last month of the lease of DS j , $j \in \hat{J}$
- y_{0jk} : initial assignment, $j, k \in J$; $y_{0jk} = 1$ if block k is initially assigned to station j ; $y_{0jk} = 0$ otherwise
- q_{tj} : number of packages to be delivered to block j at month t , $t \in T, j \in J$
- α^{cap} : constant term of station capacity-DS size function
- β^{cap} : coefficient of DS size in station capacity-DS size function
- α^{cst} : constant term of the station cost
- β_j^{cst} : unit DS size cost of block j (in RMB/month/ m^2), $j \in J$

- s_{0j} : initial DS size of block j ; if there is no initial DS at block j , $s_{0j} = 0$, $j \in J$
- s^{lb}, s^{ub} : lower bound and upper bound of size of new DSs
- a_{jk} : connectivity indicator; $a_{jk} = 1$ if block j and k are adjacent, $a_{jk} = 0$ otherwise, $j \in J$

Decision Variables

- $X_{tj} = \begin{cases} 1 & \text{if a DS exists at block } j \text{ at period } t, \\ 0 & \text{otherwise,} \end{cases} \quad t \in T \cup \{0\}, j \in J$
- $Y_{tjk} = \begin{cases} 1 & \text{if block } k \text{ is assigned to DS} \\ j \text{ at period } t, \\ 0 & \text{otherwise,} \end{cases} \quad t \in T \cup \{0\}, j, k \in J$
- $W_{tk} = \begin{cases} 1 & \text{if the station to which block } k \text{ is} \\ \text{assigned changes at period } t, \\ 0 & \text{otherwise,} \end{cases} \quad t \in T, k \in J$

- $S_{tj} \geq 0$: DS size of block j at month t , $t \in T \cup \{0\}, j \in J$
- $Z_{tj} \geq 0$: DS cost of block j at month t , $t \in T \cup \{0\}, j \in J$
- $F_{kl}^{tji} \geq 0$: flow originating from block j , passing through block k and l , which are adjacent ($a_{kl} = 1$), and terminated in block i at month t , $t \in T, i, j, k, l \in J, a_{kl} = 1$

Objective Function and Constraints

$$(P): \min \sum_{t \in T} \sum_{j \in J} Z_{tj} + \sum_{t \in T} \sum_{j \in J} \sum_{k \in J} q_{tk} (\alpha_d + \beta_d d_{jk}) Y_{tjk} \quad (A.1)$$

$$\text{s.t. } X_{0j} = 1 \quad \forall j \in \hat{J} \quad (A.2)$$

$$X_{0j} = 0 \quad \forall j \in \bar{J} \quad (A.3)$$

$$S_{0j} = s_{0j} \quad \forall j \in J \quad (A.4)$$

$$X_{tj} = 0 \quad \forall t \in T, \forall j \in \bar{J} \quad (A.5)$$

$$X_{tj} = 1 \quad \forall j \in \hat{J}, \forall t \in \{1, 2, \dots, m_j\} \quad (A.6)$$

$$S_{tj} = s_{0j} \quad \forall j \in \hat{J}, \forall t \in \{1, 2, \dots, m_j\} \quad (A.7)$$

$$X_{t,j} = X_{t-1,j} \quad \forall j \in \hat{J}, \forall t > m_j + 1 \quad (A.8)$$

$$S_{t,j} = s_{0j} X_{t,j} \quad \forall j \in \hat{J}, \forall t \geq m_j + 1 \quad (A.9)$$

$$X_{t-1,j} \leq X_{tj} \quad \forall j \in \hat{J}, \forall t \in T \quad (A.10)$$

$$S_{tj} \leq S_{t-1,j} + s^{ub} (1 - X_{t-1,j}) \quad \forall j \in \hat{J}, \forall t \in T \quad (A.11)$$

$$S_{t-1,j} \leq S_{t,j} \quad \forall j \in \hat{J}, \forall t \in T \quad (A.12)$$

$$S_{tj} \leq s^{ub} X_{t,j} \quad \forall j \in \hat{J}, \forall t \in T \quad (A.13)$$

$$S_{tj} \geq s^{lb} X_{t,j} \quad \forall j \in \hat{J}, \forall t \in T \quad (A.14)$$

$$\sum_{j \in \hat{J}} X_{tj} - \sum_{j \in \bar{J}} X_{t-1,j} \leq n_t \quad \forall t \in T \quad (A.15)$$

$$\sum_{j \in \hat{J}} X_{t-1,j} \geq \sum_{j \in \bar{J}} X_{tj} \quad \forall t \in T \quad (A.16)$$

$$Y_{0jk} = y_{0jk} \quad \forall j, k \in J \quad (A.17)$$

$$\sum_{j \in \hat{J}} Y_{tjk} = 1 \quad \forall t \in T, \forall k \in J \quad (A.18)$$

$$\sum_{k \in J} Y_{tjk} \geq X_{tj} \quad \forall t \in T, \forall j \in J \quad (A.19)$$

$$\sum_{k \in J} q_{tk} Y_{tjk} \leq \alpha^{cap} X_{tj} + \beta^{cap} S_{tj} \quad (A.20)$$

$$\forall t \in \{t^{start}, \dots, 12\}, \forall j \in J$$

$$Y_{tjk} - Y_{t-1,jk} \leq W_{tk} \quad \forall t \in T, \forall j, k \in J \quad (A.21)$$

$$\sum_{k \in J} W_{tk} \leq n_t^c \quad \forall t \in T \quad (\text{A.22})$$

$$\sum_{t \in T} W_{tk} \leq n \quad \forall k \in J \quad (\text{A.23})$$

$$\sum_{l \in J: a_{lk}=1} F_{lk}^{tji} - \sum_{l \in J: a_{kl}=1} F_{kl}^{tji} = 0 \quad \forall t \in T, \forall i, j, k \in J: j \neq i, j \neq k, i \neq k \quad (\text{A.24})$$

$$\sum_{l \in J: a_{lk}=1} F_{lk}^{tji} - \sum_{l \in J: a_{kl}=1} F_{kl}^{tji} = -Y_{tjk} \quad \forall t \in T, \forall i, j, k \in J: j \neq i, j = k \quad (\text{A.25})$$

$$\sum_{l \in J: a_{lk}=1} F_{lk}^{tji} - \sum_{l \in J: a_{kl}=1} F_{kl}^{tji} = Y_{tjk} \quad \forall t \in T, \forall i, j, k \in J: j \neq i, i = k \quad (\text{A.26})$$

$$F_{kl}^{tji} \leq Y_{tji} \quad \forall t \in T, \forall i, j, k \in J: j \neq i, a_{kl} = 1 \quad (\text{A.27})$$

$$F_{kl}^{tji} \leq Y_{tjk} \quad \forall t \in T, \forall i, j, k \in J: j \neq i, a_{kl} = 1 \quad (\text{A.28})$$

$$F_{kl}^{tji} \leq Y_{tjl} \quad \forall t \in T, \forall i, j, k \in J: j \neq i, a_{kl} = 1 \quad (\text{A.29})$$

$$Z_{tj} \geq \alpha^{cst} + \beta_j^{cst} S_{tj} - (\alpha^{cst} + \beta_j^{cst} s^{ub})(1 - X_{tj}) \quad \forall t \in T, \forall j \in J \quad (\text{A.30})$$

$$X_{tj} \in \{0, 1\} \quad \forall t \in T \cup \{0\}, \forall j \in J \quad (\text{A.31})$$

$$Y_{tjk} \in \{0, 1\} \quad \forall t \in T \cup \{0\}, \forall j, k \in J \quad (\text{A.32})$$

$$W_{tj} \in \{0, 1\} \quad \forall t \in T, \forall j \in J \quad (\text{A.33})$$

$$S_{tj} \geq 0 \quad \forall t \in T \cup \{0\}, \forall j \in J \quad (\text{A.34})$$

$$Z_{tj} \geq 0 \quad \forall t \in T, \forall j \in J \quad (\text{A.35})$$

$$F_{kl}^{tji} \geq 0 \quad \forall t \in T, \forall i, j, k, l \in J: a_{kl} = 1 \quad (\text{A.36})$$

The objective (A.1) of the model is to minimize the yearly total cost, which consists of the total fixed cost of DSs and total delivery cost from DSs to blocks. Constraints (A.2) and (A.4) initialize the locations and size of the DSs. Constraints (A.5) forbid any DSs built on blocks that cannot be selected for DS. Constraints (A.6) and (A.7) ensure that each initial delivery station cannot be canceled or change size within its lease duration. Constraints (A.8)–(A.15) guarantee that (i) each station stays open until the end of the year once it is newly opened or re-rented; (ii) once an existing DS is removed, no DS can be located on its block until the end of the year; (iii) the limits are satisfied on the number of location changes of all DSs at each period; and (iv) the size of a newly opened DS is limited within the given range. Constraint (A.16) requires that the number of DSs does not increase during the year because Qingdao has surplus DSs at the beginning and the monotonicity requirement. Constraint (A.17) initializes the assignments of blocks to DSs. Constraint (A.18) require that each block be assigned to only one DS. Constraints (A.19) and (A.20) guarantee that a DS must cover at least one block and that the capacity of each station cannot be violated. Constraints (A.21)–(A.23) ensure that the restrictions on assignment changes are satisfied. Constraints (A.24)–(A.29) ensure the connectivity among blocks covered by the same DS. Constraint (A.30) defines the fixed cost of DSs. Constraints (A.31)–(A.36) define the variables.

Appendix B. Solution Approach

It is challenging for commercial solvers to directly solve the problem (P) in a large or even medium city.

In this section, we propose a series of methods, including reformulations of connectivity constraints, valid inequalities together with a tightened formulation, and a neighborhood-based heuristics with a nested warm-start algorithm, to accelerate the solution procedure.

Reformulation of Connectivity Constraints

Enforcing connectivity among blocks served by the same DS by Constraints (A.24) to (A.29) introduces a huge number ($O(|T| \cdot |J|^4)$) of additional continuous variables F_{kl}^{tji} . We propose a reformulation of the connectivity constraints that significantly reduces the number of both decision variables and constraints. More specifically, we advise using the “aggregate” flow variables $\tilde{F}_{kl}^{tj}$, defined as the flow originating from block j , passing through two adjacent blocks k and l at month t , $t \in T, j, k, l \in J, a_{kl} = 1$. Immediately, we have $\tilde{F}_{kl}^{tj} = \sum_{i \in J: i \neq j} F_{kl}^{tji}$. Thus, Constraints (A.24) to (A.29) are equivalent to the following constraints:

$$\sum_{k \in J, a_{jk}=1} \tilde{F}_{jk}^{tj} - \sum_{k \in J, a_{kj}=1} \tilde{F}_{kj}^{tj} = \sum_{k \in J, k \neq j} Y_{tjk} \quad \forall t \in T, \forall j \in J \quad (\text{B.1})$$

$$\sum_{k \in J, a_{lk}=1} \tilde{F}_{lk}^{tj} - \sum_{k \in J, a_{kl}=1} \tilde{F}_{kl}^{tj} = -Y_{tjl} \quad \forall t \in T, \forall l \in J: l \neq j \quad (\text{B.2})$$

$$\tilde{F}_{kl}^{tj} \leq (|J| - 1)Y_{tjk} \quad \forall t \in T, \forall j, k, l \in J: a_{kl} = 1 \quad (\text{B.3})$$

$$\tilde{F}_{kl}^{tj} \leq (|J| - 1)Y_{tjl} \quad \forall t \in T, \forall j, k, l \in J: a_{kl} = 1 \quad (\text{B.4})$$

The intuition behind this reformulation is as follows: each DS discharges a certain number (which equals the number of blocks it serves) of units of flow to the blocks that it serves (Constraint (B.1)), whereas each block “absorbs” one unit of flow from its assigned DS (Constraint (B.2)); thus, the flow between any two adjacent blocks discharged by a certain DS should not be more than the total number of blocks that the DS serves minus one, which is upper-bounded by $|J| - 1$ (Constraints (B.3) and (B.4)). Note that the number of flow variables $\tilde{F}_{kl}^{tj}$ is now $O(|T| \cdot |J|^3)$ and the number of constraints are significantly reduced compared with Constraints (A.24)–(A.29).

Valid Inequalities and a Tightened Formulation

We used the following two types of valid inequalities to strengthen Problem (P):

$$Y_{tjk} \leq X_{tj} \quad \forall t \in T, \forall j, k \in J \quad (\text{B.5})$$

$$\sum_{j \in J} \alpha^{cap} X_{tj} + \sum_{j \in J} \beta^{cap} S_{tj} \geq \sum_{k \in J} q_{tk} \quad \forall t \in T \quad (\text{B.6})$$

Inequalities (B.5), which are typically used in facility location and network design problems, are well-known tight upper bounds on the block assignment variables. Inequalities (B.6) are the aggregate demand constraints and can enable mixed-integer programming solvers to generate cover cuts that further strengthen the formulation (Jena et al. 2017).

We can further tighten the formulation by replacing the number $|J| - 1$ used in Constraints (B.3) and (B.4) with smaller upper bounds $M_t = \max\{|A| : A \subseteq J, \sum_{k \in A} q_{tk} \leq s^{ub}\}$. Therefore,

Table C.1. The Computation Performance Results for the Neighborhood-Based Heuristic with the Nested Warm-Start Algorithm in JD.com Network in Dalian

Relative size, τ^a	Marginal time (seconds) ^b	Accumulative time (seconds) ^c	Heuristic gap (%)
1	6.54	6.54	5.616
2	16.68	23.22	3.620
3	627	650.22	1.721
4	3358	4,008.22	1.215
5	493	4,501.22	0.772
6	272	4,773.22	0.379
7	865	5,638.22	0.262
8	1,886	7,524.22	0.198
9	752	8,276.22	0.021
10	1,646	9,922.22	0.000
20	3,428	13,350.22	0.000
50	3,859	17,245.22	0.000
100	15,010	32,255.22	0.000

^aRelative size of restricted DS neighborhood.^bThe computation time of solving the current problem by using the optimal solution of the former problem as the warm-start solution.^cThe total computation time of solving the current problem.

we derived the following strengthened formulation:

$$\begin{aligned}
 (\mathbf{P}'): \quad & \min \quad (\text{A.1}) \\
 \text{s.t.} \quad & (\text{A.2})\text{--}(\text{A.23}), (\text{A.30})\text{--}(\text{A.35}), (\text{B.1}), (\text{B.2}), (\text{B.5}), (\text{B.6}) \\
 & \tilde{F}_{kl}^{ij} \leq M_i Y_{tjk} \quad \forall t \in T, \forall j, k, l \in J : a_{kl} = 1 \\
 & \tilde{F}_{kl}^{ij} \leq M_i Y_{tjl} \quad \forall t \in T, \forall j, k, l \in J : a_{kl} = 1 \\
 & \tilde{F}_{kl}^{ij} \geq 0 \quad \forall t \in T, \forall j, k, l \in J : a_{kl} = 1
 \end{aligned}$$

Heuristic

As introduced in the Solution section, we use a neighborhood-based heuristic to seek an approximation solution for problem $\mathbf{P}'$, which is the optimal solution of a $\tau\%$ -neighborhood-restricted problem for some fine-tuned τ . We defined the $\tau\%$ -neighborhood-restricted problem as problem $\mathbf{P}'$ subject to $Y_{tjk} = 0$ for all $t \in T, j \in J$ and $k \in J \setminus N_j^\tau$,

where N_j^τ is the set of the top $\tau\%$ nearest blocks of block j . In addition, we use a nested warm-start algorithm, the pseudo-code of which is shown in Algorithm 1, to more efficiently solve the $\tau\%$ -neighborhood-restricted problem.

Algorithm B.1 (Pseudo-Code for the Nested Warm-Start Algorithm)

Input: final relative size $\tau (\in [0, 100])$, an increasing sequence of relative sizes $0 \leq \tau_1 \leq \tau_2 \leq \dots \leq \tau_n = \tau \leq 100$

Output: optimal solution $\mathbf{X}$ for the $\tau\%$ -neighborhood-restricted problem

- 1: initialize solution $\mathbf{X} = 0$ and the feasibility indicator $\chi = 0$
- 2: **for all** $k \in \{1, 2, \dots, n\}$ **do**
- 3: let $\xi = \tau_k$
- 4: **if** $\chi = 0$ **then**

Table C.2. Comparison of TrC Between SCs and DSs and ToC Between Models Considering and Omitting TrC for Dalian in 2018

Month	Considering TrC			Omitting TrC			Relative change (%)	
	TrC (CNY)	ToC (CNY)	$\frac{\text{TrC}}{\text{ToC}}$ (%)	TrC (CNY)	ToC (CNY)	$\frac{\text{TrC}}{\text{ToC}}$ (%)	TrC	ToC
January	166,467	3,918,760	0.042	166,467	3,918,760	0.042	0.000	0.000
February	166,467	3,569,312	0.047	166,467	3,569,312	0.047	0.000	0.000
March	161,757	3,924,559	0.041	166,546	3,923,966	0.042	2.961	−0.015
April	147,451	3,564,223	0.041	157,111	3,569,504	0.044	6.551	0.148
May	138,014	3,861,449	0.036	152,514	3,874,812	0.039	10.506	0.346
June	138,014	3,937,675	0.035	152,514	3,949,458	0.039	10.506	0.299
July	133,377	3,885,658	0.034	143,007	3,893,570	0.037	7.220	0.204
August	133,374	4,118,249	0.032	143,002	4,125,799	0.035	7.219	0.183
September	147,721	4,632,874	0.032	152,492	4,627,805	0.033	3.230	−0.109
October	157,211	4,860,584	0.032	166,826	4,867,838	0.034	6.116	0.149
November	157,211	5,545,631	0.028	166,826	5,546,346	0.030	6.116	0.013
December	157,205	5,526,167	0.028	166,827	5,534,624	0.030	6.120	0.153
Total	1,804,268	51,345,141	0.035	1,900,599	51,401,794	0.037	5.339	0.110

Note. CNY, Chinese yuan; ToC, total cost; TrC, transportation cost.

Table C.3. Plan Comparison for Dalian in 2018

	Manual plan	Model plan (% change)	Benchmark plan (% change)
Total cost (CNY)	52,124,876	49,501,195 (−5.03%)	47,845,723 (−8.21%)
Fixed cost (CNY)	10,280,555	9,470,984 (−7.87%)	8,666,973 (−15.70%)
Delivery cost (CNY)	41,844,321	40,030,211 (−4.34%)	39,178,750 (−6.37%)
DS number in the last month	39	35 (−10.26%)	29 (−25.64%)
DS capacity use rate	68.30%	67.20% (−1.1 pp)	68.58% (0.28 pp)
Average delivery distance (km)	6.20	4.85 (−21.72%)	4.22 (−31.90%)

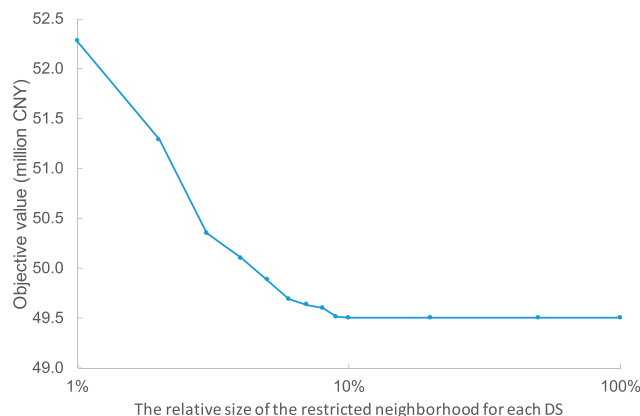
Note. CNY, Chinese yuan.

- 5: check the feasibility of the $\xi\%$ -neighborhood-restricted problem
- 6: **if** the $\xi\%$ -neighborhood-restricted problem is feasible **then**
- 7: solve the $\xi\%$ -neighborhood-restricted problem
- 8: let X = the optimal solution of the $\xi\%$ -neighborhood-restricted problem
- 9: let $\chi = 1$
- 10: **end if**
- 11: **else**
- 12: solve the $\xi\%$ -neighborhood-restricted problem with initial solution X
- 13: update X = the optimal solution of the $\xi\%$ -neighborhood-restricted problem
- 14: **end if**
- 15: **end for**

Appendix C. Supplementary Results

Here we provide a supplementary case study on the delivery network planning in Dalian. Dalian is a medium-sized Chinese city in Liaoning Province, consisting of 35 DSs and 283 blocks at the beginning of 2018. The R^2 of the linear model between the delivery cost per package and the delivery distance for Dalian is 0.896. First, we summarized the computation performance of our solution method for Dalian in Table C.1 and Figure C.1. For Dalian, we obtained the optimal solution of the

Figure C.1. (Color online) Objective Value Change in the Solution Procedure for the Dalian Instance



Note. This figure illustrates how the objective value (or the total cost) of the neighborhood-restricted problem changes as the relative size of the restricted neighborhood varies in the solution procedure for the Dalian instance.

unrestricted problem by solving the 10%-neighborhood-restricted problem within 3 hours. Second, we confirmed for Dalian the assumption of omitting the transportation cost between SCs and DSs by the results shown in Table C.2. Third, we show the comparison among the manual plan, our model plan, and the benchmark plan (without initial DS locations and block-DS assignments) for Dalian in Table C.3. The manual plan in Dalian has reached an “ideal” DS capacity use rate, which suggests that its delivery system is highly efficient. Nonetheless, our model can still save 5% of its total cost and, more significantly, reduce the average delivery distance by over 20%.

Relative size of restricted DS neighborhood. The computation time of solving the current problem by using the optimal solution of the former problem as the warm-start solution. The total computation time of solving the current problem.

References

- Aardal K (1998) Capacitated facility location: Separation algorithm and computational experience. *Math. Programming* 81:149–175.
- Aktaş E, Özyaydin Ö, Bozkaya B, Ülengin F, Önsel Ş (2013) Optimizing fire station locations for the Istanbul Metropolitan Municipality. *Interfaces* 43(3):240–255.
- Allen JD, Tobin RL, Calderan A (2017) Verizon optimizes work center locations to reduce installation and repair operations costs. *Interfaces* 47(2):111–121.
- Arabani AB, Farahani RZ (2012) Facility location dynamics: An overview of classifications and applications. *Comput. Indust. Engrg.* 62(1):408–420.
- Canel C, Khumawala BM (1997) Multi-period international facilities location: An algorithm and application. *Internat. J. Production Res.* 35(7):1891–1910.
- Castro J, Nasini S, Saldanhadagama F (2017) A cutting-plane approach for large-scale capacitated multi-period facility location using a specialized interior-point method. *Math. Programming* 163(1):411–444.
- Chardaire P, Sutter A, Costa M (1996) Solving the dynamic facility location problem. *Networks* 28(2):117–124.
- Christofides N, Beasley JE (1983) Extensions to a Lagrangean relaxation approach for the capacitated warehouse location problem. *Eur. J. Oper. Res.* 12(1):19–28.
- Daskin MS (2013) *Network and Discrete Location: Models, Algorithms, and Applications*, 2nd ed. (Wiley, Hoboken, NJ).
- Dias J, Captivo ME, Clímaco J (2007) Efficient primal-dual heuristic for a dynamic location problem. *Comput. Oper. Res.* 34(6):1800–1823.
- Duran S, Gutierrez MA, Keskinocak P (2011) Pre-positioning of emergency items for CARE International. *Interfaces* 41(3):223–237.
- Guignard M, Spielberg K (1979) A direct dual method for the mixed plant location problem with some side constraints. *Math. Programming* 17(1):198–228.

- Ho SC (2015) An iterated tabu search heuristic for the single source capacitated facility location problem. *Appl. Soft Comput.* 27:169–178.
- Jacobsen SK (1983) Heuristics for the capacitated plant location model. *European J. Oper. Res.* 12(3):253–261.
- Jena SD, Cordeau JF, Gendron B (2017) Lagrangian heuristics for large-scale dynamic facility location with generalized modular capacities. *INFORMS J. Comput.* 29(3):388–404.
- Klose A, Görtz S (2007) A branch-and-price algorithm for the capacitated facility location problem. *Eur. J. Oper. Res.* 179(3):1109–1125.
- Köksalan M, Süral H (1999) Efes Beverage Group makes location and distribution decisions for its malt plants. *Interfaces* 29(2):89–103.
- Magnanti TL, Wong RT (1981) Accelerating benders decomposition: Algorithmic enhancement and model selection criteria. *Oper. Res.* 29(3):464–484.
- Melkote S, Daskin MS (2001) Capacitated facility location/network design problems. *Eur. J. Oper. Res.* 129(3):481–495.
- Nickel S, Saldanha da Gama F (2015) Multi-period facility location. Laporte G, Nickel S, Saldanha da Gama F, eds. *Location Science* (Springer, Boston), 289–210.
- Santibáñez P, Bekiou G, Yip K (2009) Fraser Health uses mathematical programming to plan its inpatient hospital network. *Interfaces* 39(3):196–208.
- Tran TH, Scaparra MP, O'Hanley JR (2017) A hypergraph multi-exchange heuristic for the single-source capacitated facility location problem. *Eur. J. Oper. Res.* 263(1):173–187.
- van den Berg PL, Legemaate GAG, van der Mei RD (2017) Increasing the responsiveness of firefighter services by relocating base stations in Amsterdam. *Interfaces* 47(4):352–361.
- Wesolowsky GO, Truscott WG (1975) The multiperiod location-allocation problem with relocation of facilities. *Management Sci.* 22(1):57–65.
- Xie S, Chen X, Wang Z, Ouyang Y, Somani K, Huang J (2016) Integrated planning for multiple types of locomotive work facilities under location, routing, and inventory considerations. *Interfaces* 46(5):391–408.

Verification Letter

Gang Liu, Logistics Network Planner, Comprehensive Planning Team, JD.com, No. 18 Kechuang 11 Street, BDA, Beijing 101111, China, writes:

"I worked on the Comprehensive Planning Team of JD Logistics, and I was responsible for logistics network planning in 2017 and 2018. I verify that the paper titled "JD.com Improves Delivery Networks by a Multiperiod Facility Location Model" by Ningxuan Kang, Hao Shen, and Ye Xu describes a real-life optimization tool for improving the design of JD.com delivery networks.

"The mission of JD Logistics is to "shorten the fulfillment process to increase efficiency, and access customers faster with increased precision." To achieve this goal, continuous improvement of delivery networks is always on the way. The model presented in this paper, which outputs an optimal plan for delivery station location and station-area assignment, provides a mathematical method that utilizes operations research techniques to save logistics costs and increase delivery efficiency. The result shows what an optimal delivery network is like and how many costs we can economize.

"Although it is difficult to set up such networks in one step considering practical restrictions, we have integrated the model of this paper into our decision tools in network design. The results given by the model provided valuable guidance on 2018 annual planning of networks. JD.com will endeavor to provide high-quality service to customers with efficient delivery networks."

Ningxuan Kang is an applied operations research scientist at JD.com in Beijing, China. He received BS degrees in micro-electro-mechanical system engineering and economics in 2011 and a PhD degree in industrial engineering in 2016, all from Tsinghua University, Beijing, China. His research interests include analysis and optimization of production and supply chain systems.

Hao Shen is an assistant professor of operations management in the School of Business at Renmin University of China. He received his BS in engineering mechanics (Tsien Hsue-Shen Class) and his PhD in industrial engineering, both from Tsinghua University. His research focuses on the application of operations research and data analytics in supply chain management. He has publications in journals such as *Manufacturing & Service Operations Management* and *Production and Operations Management*.

Ye Xu is a principal research manager at Bianlifeng, a Chinese technological innovation retail enterprise. Prior to joining Bianlifeng, she worked for JD.com, leading a team of operations researchers to work on supply chain management projects, including the one introduced in this paper. She has also worked for Google and Amazon. She received a BS from Tsinghua University, and an MS and a PhD in industrial engineering and operations research, respectively, from the University of California, Berkeley.